



GTP Programme – Bootcamp Guide (2026 Edition)

1. Programme Overview

The **Graduate Training Programme (GTP) 2026** is a structured, production-oriented bootcamp designed to develop industry-ready full-stack engineers and quality professionals.

Participants will build, test, deploy, and productionize a complete **Project Management Tool** using a modern 2026 enterprise technology stack aligned with real-world delivery standards.

2. Programme Essentials

2.1 Design & Collaboration Tools

- **Figma** (Latest Web Version 2026)
- **GitHub** Account (with SSH key setup)
- Professional Email + WhatsApp (team communication)
- **Jira** (Project & Test Management)

2.2 Coding Practice Platforms

- **HackerRank**
 - SQL (Basic & Intermediate)
 - Problem Solving in C#

2.3 Hardware Requirements

Component	Minimum	Recommended
RAM	8 GB	16 GB
Storage	256 GB SSD	512 GB SSD
Processor	i5 / Ryzen 5	i7 / Ryzen 7

SSD storage is mandatory for smooth development performance.

3. 2026 Core Technology Stack

3.1 Backend Stack

- .NET 8 (LTS)
- ASP.NET Core Web API (.NET 8)
- Entity Framework Core 8
- Dapper 2.x
- HotChocolate GraphQL 14+
- SQL Server 2022
- PostgreSQL 16

3.2 Frontend Stack

- React 18.3+
- Vite 5+
- Material UI v6
- Apollo Client
- Axios
- TanStack Query (React Query 5)

3.3 Mobile Stack

- Flutter 3.19+
- Dart 3
- GraphQL Flutter

3.4 DevOps & Deployment

- Docker 26+
- Docker Compose v2
- Vercel

- GitHub Actions
- Azure / Railway / Render

4. Programme Structure (7 Weeks Intensive)

Week 1: Development Fundamentals

Goals

- Complete all required HackerRank challenges
- Set up a production-grade development environment

Installation & Setup

- .NET 8 SDK
- SQL Server 2022 Developer Edition
- PostgreSQL 16
- DBeaver 24+
- Visual Studio 2022
- VS Code
- Node.js 20 LTS
- Docker Desktop

Week 2: Backend & API Development

Capstone Project: Project Management Tool

Core Deliverables

- Database schema design in Figma
- Implementation in SQL Server
- Stored procedures & optimized queries
- REST APIs using .NET 8
- GraphQL layer via HotChocolate
- Bulk operations & advanced data handling
- API testing using Postman

Week 3: Frontend Development (React 18 + Vite)

Key Deliverables

- React app setup with Vite
- Material UI v6 integration
- REST integration (Axios + TanStack Query)
- GraphQL integration (Apollo Client)
- Secure authentication & state management

Week 4: Mobile Development (Flutter 3.19+)

Deliverables

- Complete login & authentication screens
- REST API integration
- GraphQL endpoint integration
- Advanced state management (Provider / Riverpod)
- Fully responsive cross-device UI

Week 5: Deployment & DevOps

Frontend Deployment (Vercel)

1. Push project to GitHub
2. Import repository into Vercel
3. Select Vite framework preset
4. Configure environment variables
5. Deploy & verify live URL

Backend Dockerization

- Create production-ready Dockerfile
- Build image
- Run container

Docker Compose

- Create docker-compose.yml (API + SQL Server)
- Build and orchestrate services

Optional Advanced Deployment

- Push image to Docker Hub
- Deploy to Azure / Railway / Render
- Implement GitHub Actions CI/CD

Week 6: Quality Engineering Training

Goals

- Master professional Quality Engineering principles
- Implement enterprise-grade testing strategies

Step-by-Step

1. Quality Engineering Fundamentals

- SDLC vs STLC
- Testing types (Unit, Integration, System, Regression, UAT)
- KPIs & Quality Metrics

2. Manual Testing Excellence

- Test cases & scenarios
- Exploratory testing
- Defect severity & priority

3. Test Management & Documentation

- Full Jira workflow
- Traceability matrices

4. API Testing Mastery

- Advanced Postman collections
- Environment variables
- Assertions

- Newman CLI automation
- Contract testing
- 5. **Unit & Integration Testing**
 - xUnit / NUnit (.NET)
 - Jest + React Testing Library
 - Flutter widget & integration tests
- 6. **Automated UI Testing**
 - Selenium / Cypress (React)
 - Flutter integration testing
- 7. **Non-Functional Testing**
 - Performance testing fundamentals
 - OWASP Top 10 security basics
 - Accessibility testing
- 8. **DevOps Integration & TDD**
 - Test-Driven Development
 - Minimum 80% code coverage
 - CI/CD test embedding

Week 7: Real-World Quality Engineering Practice

Goals

- Apply QE skills in a live production environment
- Deliver production-grade quality assurance

Practical Responsibilities

- Complete test strategy & plans for the capstone
- Execute full manual testing cycles
- Develop automated test suites (≥80% coverage)
- Log & track defects in Jira with evidence
- Perform QE on Sababisha Solutions' internal products
- Conduct performance benchmarking & UAT

- Produce executive test summary reports
- Integrate automated testing into CI/CD pipelines

5. Programme Outcomes

Upon successful completion, participants will:

- Build and deploy a production-ready full-stack system
- Implement enterprise-level Quality Engineering practices
- Achieve CI/CD-integrated automated testing
- Be deployment-ready for Sababisha Solutions delivery teams

Welcome to the 2026 Cohort

Engineering Excellence. Production Mindset. Professional Impact.